

Exercise 1 PL/SQL – Sanskar Kumawat

Part 0 – Created a Database Table

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'exercise_1_pl/sql' selected. The main editor shows the following SQL script:

```
1 CREATE TABLE Customers (  
2   CustomerID INTEGER PRIMARY KEY, name VARCHAR(50), Age INTEGER,  
3   Balance DECIMAL(10, 2), VIP_Member VARCHAR(5) DEFAULT 'FALSE'  
4 );  
5 CREATE TABLE Loans (  
6   LoanID INT PRIMARY KEY, CustomerID INT, InterestRate DECIMAL(5,2), DueDate DATE,  
7   FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)  
8 );  
9 INSERT INTO Customers VALUES  
10 (1, 'abhilash', 65, 13000, 'FALSE'),  
11 (2, 'Priyash', 45, 7000, 'FALSE'),  
12 (3, 'Amit', 70, 15000, 'FALSE'),  
13 (4, 'Sneha', 30, 9000, 'FALSE'),  
14 (5, 'Rohan', 62, 20000, 'FALSE');  
15  
16 INSERT INTO Loans VALUES  
17 (101, 1, 8.50, '2026-07-30'),  
18 (102, 2, 9.00, '2026-09-14'),  
19 (103, 3, 7.80, '2026-07-15'),  
20 (104, 4, 8.75, '2026-08-25'),  
21 (105, 5, 9.20, '2026-07-18');
```

The bottom panel shows the 'Output' tab with the following results:

#	Time	Action	Message
1	16:55:39	CREATE TABLE Customers (CustomerID INTEGER PRIMARY KEY, name VARCHAR(50), Age INTEGER, Bal...	0 row(s) affected
2	16:57:00	CREATE TABLE Loans (LoanID INT PRIMARY KEY, CustomerID INT, InterestRate DECIMAL(5,2), DueDate DA...	0 row(s) affected
3	16:59:07	INSERT INTO Customers VALUES (1,'abhilash',65,13000,'FALSE'), (2,'Priyash',45,7000,'FALSE'), (3,'Amit',70,150...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
4	16:59:07	INSERT INTO Loans VALUES (101,1,8.50,'2026-07-30'), (102,2,9.00,'2026-09-14'), (103,3,7.80,'2026-07-15'), (10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

	CustomerID	name	Age	Balance	VIP_Member		LoanID	CustomerID	InterestRate	DueDate
▶	1	abhilash	65	13000.00	FALSE	▶	101	1	8.50	2026-07-30
	2	Priyash	45	7000.00	FALSE		102	2	9.00	2026-09-14
	3	Amit	70	15000.00	FALSE		103	3	7.80	2026-07-15
	4	Sneha	30	9000.00	FALSE		104	4	8.75	2026-08-25
	5	Rohan	62	20000.00	FALSE		105	5	9.20	2026-07-18
*	NULL	NULL	NULL	NULL	NULL	*	NULL	NULL	NULL	NULL

Part 1 – scenario 1 - The bank wants to apply a discount to loan interest rates for customers above 60 years old.(1% reduction)

The screenshot shows the MySQL Workbench interface with the following SQL script:

```
22  
23 • UPDATE Loans  
24   SET InterestRate = InterestRate - 1  
25   WHERE CustomerID IN (  
26     SELECT CustomerID  
27     FROM Customers  
28     WHERE Age > 60  
29   );  
30 • SELECT * FROM Loans
```

	LoanID	CustomerID	InterestRate	DueDate
▶	101	1	6.50	2026-07-30
	102	2	9.00	2026-09-14
	103	3	5.80	2026-07-15
	104	4	8.75	2026-08-25
	105	5	7.20	2026-07-18
*	NULL	NULL	NULL	NULL

output

Part 2 - scenario 2 : A customer can be promoted to VIP status based on their balance.

```

29  );
30  • SET SQL_SAFE_UPDATES = 0;
31
32  • UPDATE Customers
33      SET VIP_Member = 'TRUE'
34      WHERE Balance > 10000;

```

CustomerID	name	Age	Balance	VIP_Member
▶ 1	abhilash	65	13000.00	TRUE
2	Priyash	45	7000.00	FALSE
3	Amit	70	15000.00	TRUE
4	Sneha	30	9000.00	FALSE
5	Rohan	62	20000.00	TRUE
*	NULL	NULL	NULL	NULL

Part 3 - scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.

```

36
37  • SELECT
38      c.name,
39      l.LoanID,
40      l.DueDate
41  FROM Customers c
42  JOIN Loans l
43  ON c.CustomerID = l.CustomerID
44  WHERE l.DueDate BETWEEN CURDATE() AND DATE_ADD(CURDATE(), INTERVAL 30 DAY);

```

name	LoanID	DueDate
▶ abhilash	101	2026-07-30
Amit	103	2026-07-15
Rohan	105	2026-07-18